# Coen Adler

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**Email**: ctadler@uci.edu
**GitHub**: Coaster41
**Website**: coaster41.github.io
**Phone**: (510) 314-9393
**LinkedIn**: coen-adler
**Location**: Irvine, CA


## Research interests

Mechanistic Interpretability, eXplainable AI (XAI), Time Series and Tabular Foundation Models, Anomaly Detection

## Education

**University of California, Irvine** — Irvine, California
PhD in Computer Science — Sep 2024 – June 2029
Advisor: Professor Padhraic Smyth. — *GPA: 3.97*

**University of California, Santa Cruz** — Santa Cruz, California
BS in Computer Science — Sep 2021 – June 2024
Advisors: Professors Leilani Gilpin, Luca de Alfaro. — *GPA: 3.94*

## Papers

**Beyond Accuracy: Are Time Series Foundation Models Well-Calibrated?**
**Coen Adler**, Yuxin Chang, Felix Draxler, Samar Abdi, Padhraic Smyth
*ICLR Conference, 2026.*

**Calibration Properties of Time Series Foundation Models: An Empirical Analysis**
**Coen Adler**, Yuxin Chang, Samar Abdi, Padhraic Smyth
*ICML Workshop on Foundation Models for Structured Data, 2025.*

**Model Pruning and Explainability: An Experience Report**
**Coen Adler**, Leilani Gilpin, Tyler Sorenson
*UC Santa Cruz - Undergraduate Thesis, 2024.*

**Diffusion Models for Species-Specific Functional Habitat Connectivity**
Natalia Ocampo-Peñuela, Luca de Alfaro, **Coen Adler**, et al.
*UCSC Technical Report - Preprint, under review. Invited talk, Google Geo For Good Summit, 2023.*

**Fair Augmentation of Decision Trees Through Selective Node Retraining**
**Coen Adler**, Leilani Gilpin.
*BayLearn Symposium - Abstract, poster presentation, 2023.*

**Utilizing Segment Anything Model for Assessing Localization of GRAD-CAM in Medical Imaging**
Evan Kellener, Ihina Nath, An Ngo, Thomas Nguyen, **Coen Adler**, et al.
*arXiv, 2023.*

**Analysis of Training PINNs on Multiple Computational Fluid Dynamics Models Consisting of Different Shapes**
Bryce Montgomery, Kevin Bachelor, Manju Shettar, **Coen Adler**, et al.
*Unpublished Manuscript, 2023.*

## Research Labs

**The Datalab**
Advisor: Padhraic Smyth (UCI) — September 2024 – Present
Time series and tabular foundation models: mechanistic interpretability and calibration of TSFMs using sparse autoencoders, anomaly detection in large-scale univariate time series with Google using SHAP, Integrated Gradients, and GradCAM to explain false positives, and evaluation of tabular foundation models against fine-tuned LLM baselines.

**ACM Undergraduate AI Research Lab**
Advisors: **Coen Adler**, Arnav Kartikeya (UCSC) — March 2023 – June 2024
Undergraduate AI-focused research lab with the goal of introducing AI research to passionate undergraduate students: Usage of Meta's Segment Anything Model on GRAD-CAM for medical segmentation, Applications of Physics-Informed Neural Networks on complex geometric aerodynamics.

**Artificial Intelligence Accountability Explainability Lab**
Advisor: Professor Leilani Gilpin (UCSC) — December 2021 – June 2024
Explainable AI with work in robust and fair AI systems: GAN applications in AV robustness, Benchmarking disagreements of SHAP and LIME explanations, Fair augmentation of Decision Trees. Led lab organization and resource/computing usage.

**Computational Ecology Lab**
Advisor: Professor Luca de Alfaro (UCSC) — December 2021 – June 2024
Applications of game theory and machine learning to the study of interactions: Diffusion model for mapping forest bird habitat connectivity, Architecture for bird mapping browser tool.

## Industry experience

**Cisco Systems,** Secure Equipment Access — San Jose, California
*Software Engineering Intern II* — Summer 2024

Extensible Golang asset connection workflow engine for device health and security monitoring using Splunk and Ollama, reduced batch computing overhead through multi-threaded integration in REST API server and SSH client engine.

**Cisco Systems,** Internet of Things — San Jose, California
*Software Engineering Intern I* — Summer 2023
IoT troubleshooting dashboard and architecture development: Implemented application manager microservices for accessible troubleshooting platform, Applied Apache Freemarker templating enabling runtime workflow modification, Developed REST API workflow for edge device communication.

**Cisco Systems,** Webex — San Jose, California
*Software Engineering Intern I* — Summer 2022
K8s and Docker DevOps for service cost reduction: Facilitated transition to AWS Spot EC2 and Graviton ARM Instances, Prototyped automatic node shutdown, cluster scaling, and Helm-based app installer.

## Student Leadership

**UCSC ACM Chapter,** *President* — Sep 2022 – June 2024
UCSC's largest CSE student organization, 1,000+ members. Co-founded the undergraduate AI research lab and organized weekly technical workshops and the annual hackathon.

**Santa Cruz Artificial Intelligence,** *Director of Instruction* — Sep 2022 – June 2024
Designed the AI/ML curriculum and led weekly workshops for UCSC's student AI/ML organization.

**Google Developer Student Club,** *Director of Instruction* — Sep 2022 – June 2024
Developed and taught weekly coding workshops on Google Cloud Platform, TensorFlow, and Kubernetes for UCSC's GDSC chapter.

## Honors and scholarships

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| UC Irvine Computer Science Department Research Fellowship (UCI) | 2024 |
| Chancellor's Undergraduate Research Award (UCSC) | 2024 |
| Dean's Undergraduate Research Award (UCSC) | 2024 |
| Dean's Award (UCSC) | 2021 – 2024 |

## Skills

**Tools**: Git, Kubernetes, Slurm, Docker, PostgreSQL, OpenAPI, GCP, AWS, Gradle, Helm, Neo4j
**Frameworks**: PyTorch, TensorFlow, SpringBoot, Pandas, NumPy, Ollama, Scikit-learn
**Languages**: Python, Golang, Java, C/C++, SQL, R, Flutter/Dart